Nested Bundling

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Nested Bundling

Nested bundling: More expensive bundles include all the goods of the less expensive ones.

Main Results

This paper studies when nested bundling is optimal under the assumption that (i) consumers differ in one dimension, and (ii) local quasi-concavity holds.

- Nested bundling is optimal under the *nesting condition*.

Partial Order on Bundles

- The *sold-alone quantity* $Q(b)$ is the unique quantity that maximizes Seller's revenue when he only offers b.
- Inspired by Ghili (2023), define a partial order on the set of bundles:

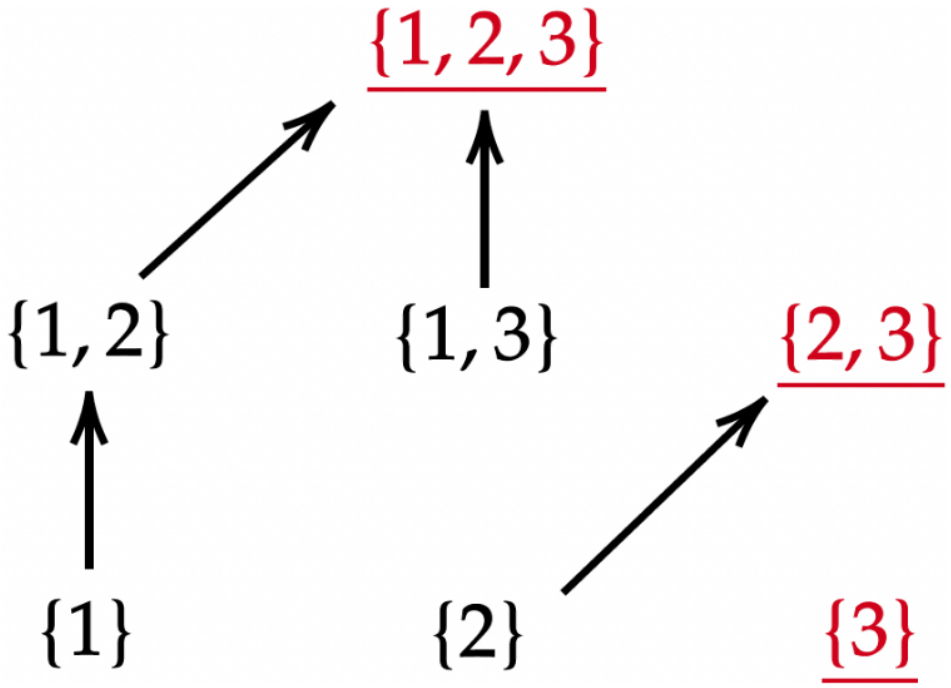
$$b_1 \preceq b_2 \iff b_1 \subseteq b_2 \text{ and } Q(b_1) \leq Q(b_2).$$

A bundle b is *dominated* if there exists $b' \neq b$ such that $b \preceq b'$ and *undominated* otherwise.

Nesting Condition

- The *nesting condition* holds if the undominated bundles can be totally ordered by set inclusion: for any two bundles b and b',

$$\text{both } b \text{ and } b' \text{ are undominated} \;\Rightarrow\; \text{either } b \subseteq b' \text{ or } b' \subseteq b. \qquad \text{(Nesting Condition)}$$



1 Introduction

2 Intuition

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Setup

- A monopolistic seller of products $\{1, \ldots, n\}$
  - $\mathcal{B} = 2^{\{1,2,\cdots,n\}}$ is the set of all bundles.
  - $C(b)$ is the production cost for bundle b.
  - A *menu* B is a set of bundles.
- A mass 1 of buyers who need at most 1 unit of each product
  - Buyer's utility $v(b, t) - p$ from consuming b and paying p
    - Privately known type $t \in \mathcal{T} = [\underline{t}, \bar{t}]$, with distribution F and continuous $f > 0$
    - Value $v(b, t) \geq 0$ for each bundle $b \in \mathcal{B}$, $v(\varnothing, t) = 0$
    - Non-additive values
  - For randomized assignment $a \in \Delta(B)$, $v(a, t) = \mathbb{E}_{b \sim a}[v(b, t)]$.
  - Assume efficiency for the highest type $\bar{t}$ to buy $\bar{b}$; i.e.,

$$\arg\max_b \{v(b, \bar{t}) - C(b)\} = \bar{b}.$$

Mechanism

- A *(stochastic, direct) mechanism* is a measurable map $(a, p) : \mathcal{T} \to \Delta\mathcal{B} \times \mathbb{R}$ that satisfies the IC and IR conditions.

$$v(a(t), t) - p(t) \geq v(a(t'), t) - p(t'), \quad \forall t, t' \in \mathcal{T}. \tag{IC}$$

$$v(a(t), t) - p(t) \geq 0, \quad \forall t \in \mathcal{T}. \tag{IR}$$

- Two mechanisms are *equivalent* if they differ on a zero-measure type set.

Consumer Types and Values

- The value function $v(b, t)$ is (i) nondecreasing in b in the set-inclusion order; (ii) continuously differentiable in t; (iii) strictly increasing in t whenever $v(b, t) > 0$.

Assumption 1 (Incremental Monotonicity)

For any two nested bundles $b_1 \subset b_2$, the incremental value $v(b_2, t) - v(b_1, t)$ is strictly increasing in t whenever it is strictly positive.

Why One-Dimensional Type Space Matters?

Multi-dimensional Screening Sucks

Why? Greater Complexity!

economic answer: inclusion, exclusion, extraction & correlated value distribution

mathematical answer: IC analysis & optimize multiple functional

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Why One-Dimensional Type Space Matters?

Make Life Easier

various ways to reduce dimensions ⇒ everybody is happy with one-dimensional thing

- distributional assumptions sufficient for some certain bundling policy

  Haghpanah and Hartline, 2021; Ghili, 2023; Frank Yang; ...

- robustness

  Carroll, 2017; Deb and Roesler, 2023; Che and Zhong, 2024; ...

- convergence rate

  Hart and Nisan, 2017; Cai, Devanur, and Weinberg, 2019; Frick, Ijima, and Ishii, 2024 ; ...

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Consumer Types and Values

- Key features of 1D type space:
  1. We can arrange consumers along a single quantity axis, with consumers positioned toward the right end having lower values for all the bundles.
     - Given a menu, they optimally choose their options.
     - Allocation rule $b(q)$ is the bundle choice for the consumer at quantity q.
     - Let $CS(q)$ denote the surplus of the consumer at q, and suppose $CS(1) = 0$.
  2. We can compute the total revenue from any feasible allocation, induced by some menu, from the *sold-alone marginal revenue curve*.
     - Let $P(b, q)$ denote the sold-alone demand curve for bundle b.
     - Let $MR(b, q)$ denotes the corresponding marginal revenue curve.

Total Revenue from Marginal Revenue

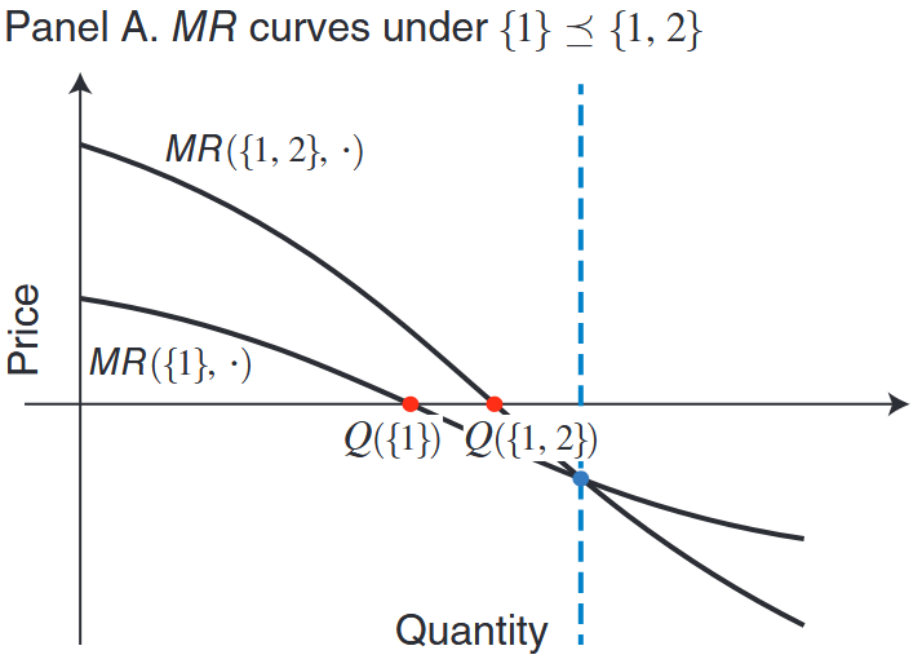
- The total revenue is

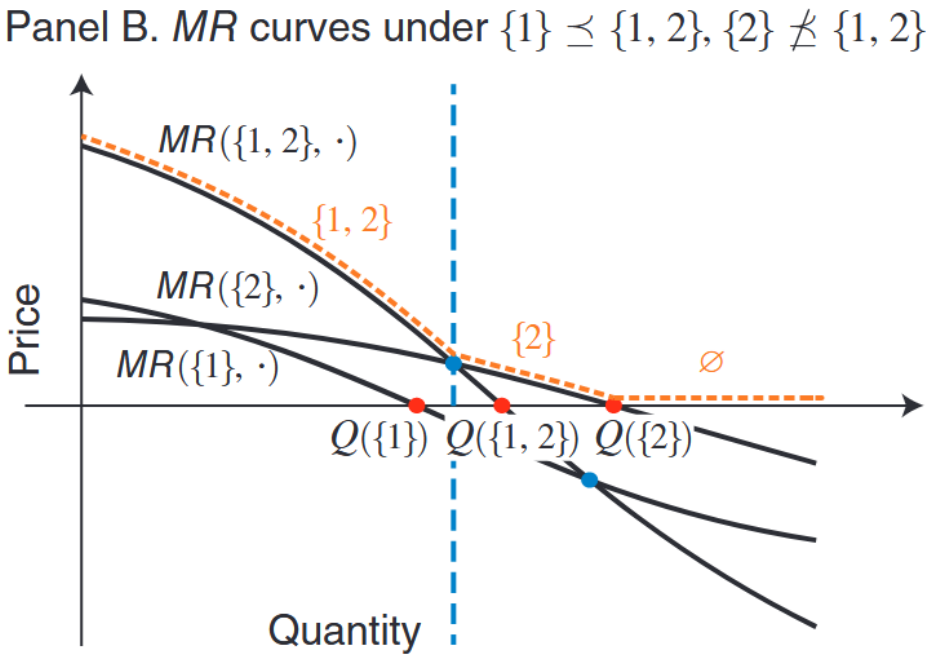
$$
\begin{aligned}
TR &= \int_0^1 P(b(q), q)\mathrm{d}q - \int_0^1 CS(q)\mathrm{d}q && \text{(Value} - \text{Consumer Surplus)} \\
&= \int_0^1 P(b(q), q)\mathrm{d}q + \int_0^1 q \cdot CS'(q)\mathrm{d}q && \text{(Integration by Parts)} \\
&= \int_0^1 P(b(q), q)\mathrm{d}q + \int_0^1 q \cdot P'_q(b(q), q)\mathrm{d}q && \text{(Envelope Theorem)} \\
&= \int_0^1 MR(b(q), q)\mathrm{d}q.
\end{aligned}
$$

Intuition (A Two-Item Case)

- Suppose we have two items $\{1, 2\}$ and costs are 0.
  - Suppose their sold-alone quantities satisfy $Q(\{1\}) < Q(\{1, 2\}) < Q(\{2\})$.
    - $\{1\} \preceq \{1, 2\}$, but $\{2\}$ is not dominated by $\{1, 2\}$.
    - The two undominated bundles are nested.
- Suppose the revenue function from selling any incremental bundle (i.e., the option to upgrade from a smaller bundle to a larger one) is strictly quasi-concave.
  - Geometric interpretation:
    - The MR curves cross zero once from above.
    - The MR curve of a larger bundle also crosses the MR curve of a smaller bundle at most once from above.

Marginal Revenue Curves





Observations

- Observations:
  1. Because $Q(\{1\}) < Q(\{1,2\})$, the two quantities must be in the region where the MR of upgrading consumers from $\{1\}$ to $\{1,2\}$ is *positive*.
     - If it is profitable to sell a consumer $\{1\}$, which happens before $Q(\{1\})$, it is even more profitable to *upgrade* the consumer to $\{1,2\}$.
  2. With the upgrading (and downgrading), we can attain the upper envelope of the MR curves by allocating bundles to consumers in a *monotone* fashion.
     - A higher-type consumer receives a larger bundle in the set-inclusion order.
     - We implement this using upgrading as follows: (1) Set the price of $\{2\}$ to be its sold-alone monopoly price; (2) Set the price of upgrading from $\{2\}$ to $\{1,2\}$ to be such that the consumer located at the threshold is indifferent to upgrading.

Local Quasiconcavity

- Under assumption (A1), for any two nested bundles $b_1 \subset b_2$, the difference $\pi(b_2, q) - \pi(b_1, q)$ is the demand curve generated by the incremental values for b_2 given b_1.
  - $\pi(b_2, q) - \pi(b_1, q)$ is the profit function of a monopolist optimizing the quantity of the incremental bundle $b_2 \setminus b_1$, given the plan of selling b_1.

Assumption 2 (Local Quasiconcavity)

For any two nested bundles $b_1 \subset b_2$, the incremental profit $\pi(b_2, q) - \pi(b_1, q)$ is strictly quasiconcave in $q \in [0, \min\{Q(b_1), Q(b_2)\}]$.

Nested bundling is optimal if there exists an optimal and nested menu.

- A menu B is *optimal*, if there exists an optimal mechanism (a, p) such that $a(t) \in B, \forall t \in \mathcal{T}$.
- A menu B is *minimal optimal*, if B is optimal but any menu $B' \subset B$ is not.
- A menu B is *nested*, if the bundles in B can be totally ordered by $\subset$.

When is Nested Bundling Optimal?

Theorem 1

Suppose that assumptions (A1) and (A2) hold. Then, under the nesting condition,

(i) Nested bundling is optimal.

(ii) A menu of undominated bundles is optimal.

(iii) Every optimal mechanism is equivalent to nested bundling.

- A sufficient and more testable condition:

$$\text{For all } b_1 \text{ and } b_2,\ Q(b_1 \cup b_2) \geq \min\{Q(b_1), Q(b_2)\}$$

(Union Quantity Condition)

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Characterization of Minimal Optimal Menus

- Let $Q(b_2|b_1)$ denote the quantity at which $\pi(b_2, q) - \pi(b_1, q)$ reaches its maximum in the interval $[0, \max\{Q(b_1), Q(b_2)\}]$.

Proposition 1

Suppose that assumptions (A1) and (A2) hold. Then, for any optimal and nested menu $B = \{b_1, \ldots, b_m\}$, where $b_1 \subset \cdots \subset b_m$, let

$$D := \{b_j \in B : Q(b_{j+1}|b_j) \geq Q(b_j|b_{j-1})\}.$$

Then menu $\tilde{B} := B \setminus D$ is also an optimal menu. If $D = \varnothing$ and $Q(b_m|b_{m-1}) > 0$, then menu B is minimal optimal.

Characterization of Minimal Optimal Menus

The following *sieve algorithm* determines the minimal optimal menu.

1. Remove all dominated bundles.
2. Remove all bundles in D.
3. Repeat Step 2 until $D = \varnothing$.

Partial Converse

- Let $b^\star$ denote the *best-selling bundle* (assumed unique); i.e.,

$$Q(b^\star) \geq Q(b), \quad \forall b \in \mathcal{B}.$$

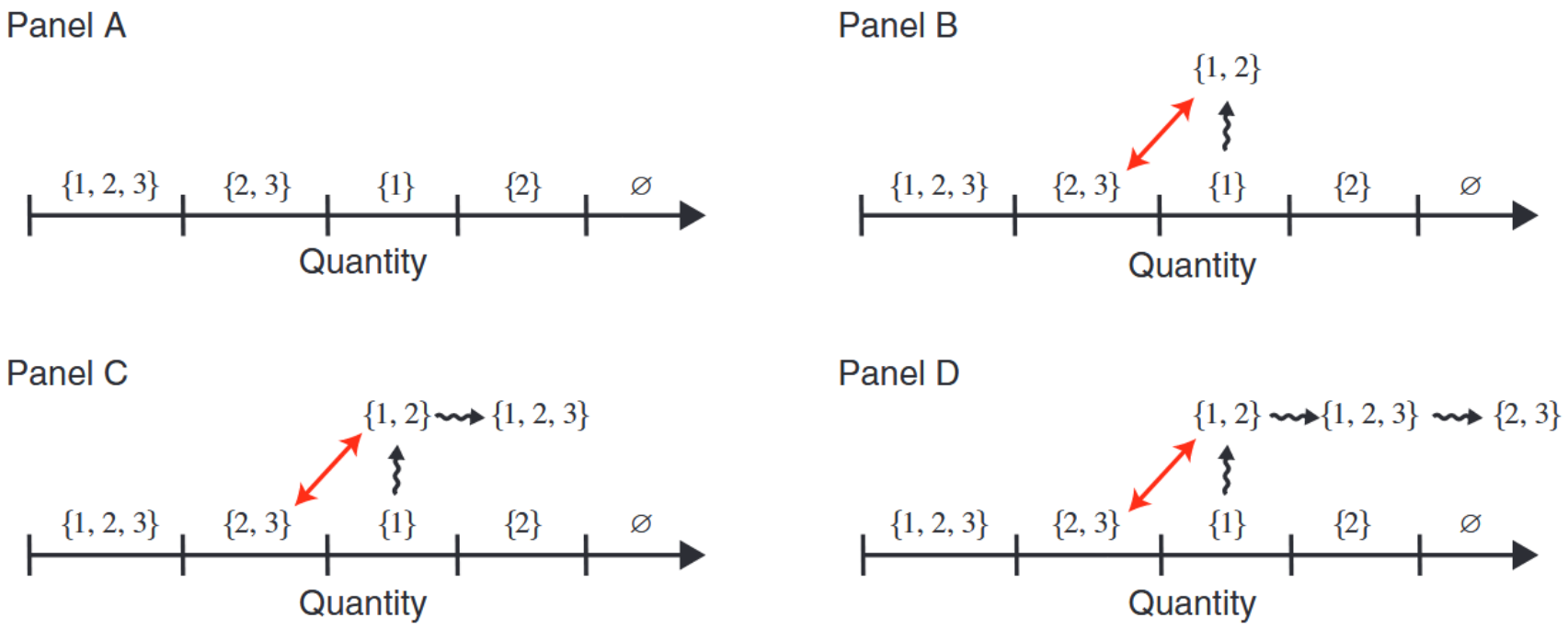
Proposition 2

Suppose that assumptions (A1) and (A2) hold. For every minimal optimal and nested menu $B = \{b_1, \ldots, b_m\}$, where $b_1 \subset \cdots \subset b_m$, we have

(1) $b_1 = b^\star$ and $b_m = \bar{b}$;

(2) $Q(b_i) > Q(b_j)$ for all $b_i \subset b_j \in B$.

Beyond Two-Item Cases



Mathematical Details

- *Monotone Construction Theorem*
  - A (mathematical) theorem with constructive proof that characterizes the range of the maximizers across parameters
  - With lattice structure imposed, this turns out a comparative statics result that generalizes Milgrom and Shannon (1994).

Mathematical Details

- Let $(X, \leq)$ be a finite partially ordered set.
  - $g : \mathcal{X} \times [0,1] \to \mathbb{R}$ satisfies *the strict single-crossing property*; i.e., for any $x_1 < x_2$ and $t < t'$,

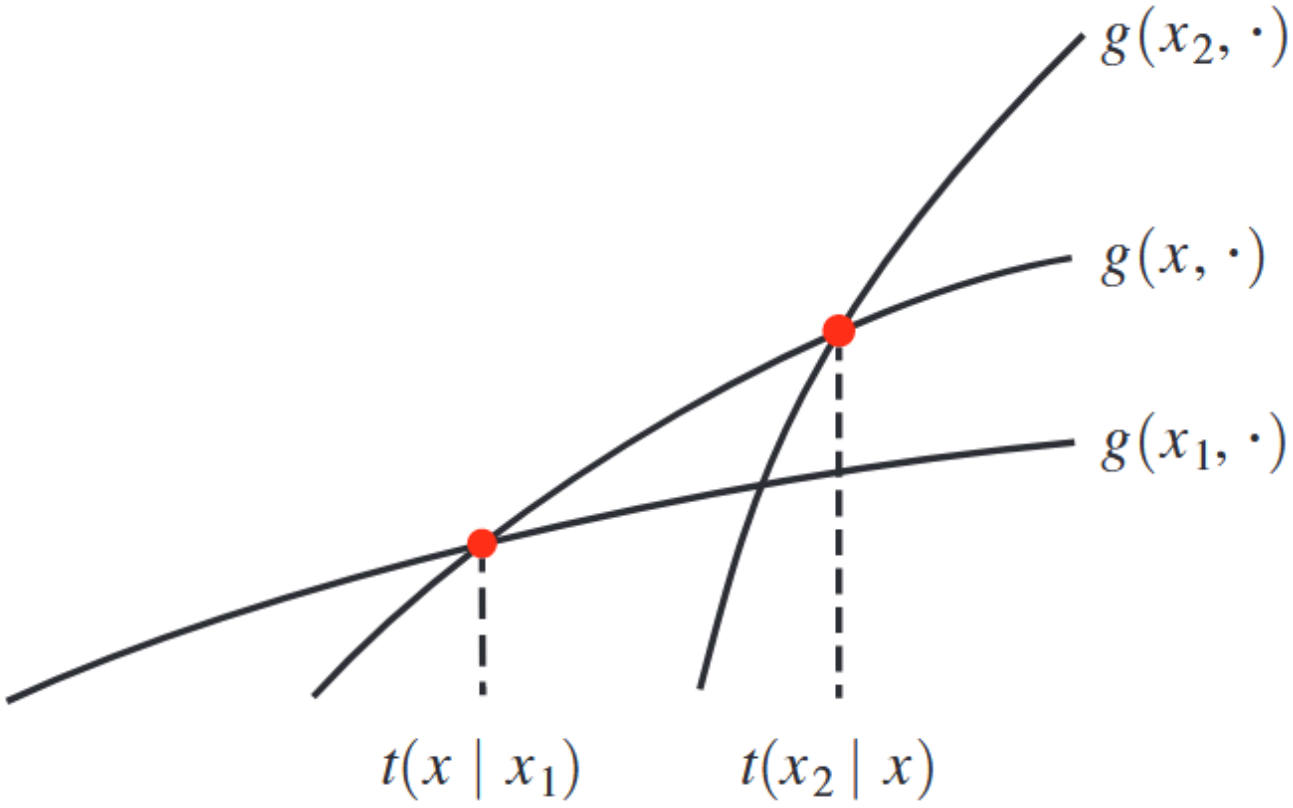
$$g(x_1, t) \leq g(x_2, t) \implies g(x_1, t') < g(x_2, t').$$

  - For any $x_1 < x_2$, let $t(x_2|x_1)$ be the unique *crossing point* of $g(x_1, t)$ and $g(x_2, t)$:

$$t(x_2|x_1) := \inf\{t \in [0,1] : g(x_1, t) < g(x_2, t)\},$$

    and put $t(x_2|x_1) = 1$ when the above set is empty.
  - An element $x \in \mathcal{X}$ is *chain-essential* for g if for all $x_1, x_2 \in \mathcal{X}$ such that $x_1 < x < x_2$, we have $t(x|x_1) < t(x_2|x)$.
  - A chain-essential element x maximizes the objective over all possible *chains* that contain x, for at least some parameter.

Chain-essential Elements



Monotone Construction Theorem

Theorem 2 (Monotone Construction Theorem)

Let $(X, \leq)$ be a finite partially ordered set. Suppose that $g : \mathcal{X} \times [0,1] \to \mathbb{R}$ is continuous in t and satisfies the strict single-crossing property in (x, t). Let $\mathcal{Y} \subseteq \mathcal{X}$ be the set of chain-essential elements for g. If $\mathcal{Y}$ is totally ordered, then there exists $x(t)$ such that

(i) $x(t) \in \operatorname{argmax}_{x \in \mathcal{X}} g(x, t)$ for all t, and $x(t)$ is the unique maximizer for almost all t;

(ii) $x(t)$ is monotone in t;

(iii) $\{x(t)\}_{t \in [0,1]} = \mathcal{Y}$.

Applications

This presentation omits the following applications of the main results.

1. Another sufficient condition for nesting characterized by elasticity
2. Some new insights into quality discrimination in Mussa and Rosen (1978)
3. Some new insights into demand rotations in Johnson and Myatt (2006) and product line design in Johnson and Myatt (2003)
4. A little extension to costly screening

Conclusion

- This paper introduces a partial order on the set of bundles defined by (i) set inclusion and (ii) sold-alone quantity.
- Nested bundling, in particular a menu of undominated bundles, is optimal if the set of undominated bundles is nested.
- The proof technique involves a new comparative statics result.